

CSE201, Fall 2025
Analysis of Algorithms
University of California, Santa Cruz

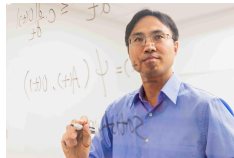
TR 8:00-9:35 am at J Bask Aud 101
09/25/25 - 12/08/25

Bookmarks

- [Personnel & Office Hours](#)
- [Course Overview](#)
- [Schedule](#)
- [Homework](#)
- [FAQ](#)

Personnel & Office Hours

Instructor: Sungjin Im (sim9@ucsc.edu). [Homepage](#)



Teaching Assistants

Qingyun Chen	Davan Harrison
qchen161@ucsc.edu	vharriso@ucsc.edu
	

Office Hour of the Instructor:

Sungjin Im: T 9:45am-10:45, 541B, Building E2 or [zoom](#).

OH of TAs/Tutors:

- Qingyun Chen: TBD
Zoom:
- Davan Harrison: TBD
Zoom:

For any non-confidential questions, please use Piazza: <https://piazza.com/ucsc/fall2025/cse201>

Course Overview

Description:

Rigorous analysis of the time and space requirements of important algorithms, including worst case, average case, and amortized analysis. Techniques include order-notation, recurrence relations, information-theoretic lower bounds, adversary arguments. Analysis of the key data structures: trees, hash tables, balanced tree schemes, priority queues, Fibonacci and binomial heaps. Algorithmic paradigms such as divide and conquer, dynamic programming, union-find with path compression, augmenting paths. Selected advanced algorithms. Introduction to NP-completeness. This course was formerly Computer Science 201.

Enrollment is restricted to graduate students; undergraduate students may enroll in this course if they have completed CSE 102 or CSE 106 and have the consent of the instructor.

Books:

- (1) CLRS book 3rd edition (textbook to be loosely followed); 4th edition is also fine.
- (2) [Kleinberg/Tardos book free online access](#)
- (3) [Tim Roughgarden's Algorithms Illuminated from Amazon](#)

Grading (Goes up to 105%; 5% is Bonus):

7 Homeworks	15% * 6 = 90% The lowest score will be dropped
Final exam	15%

* A: 90 or higher; A-: 80 or higher; B+: 70 or higher; B: 50 or higher.

If for any case, the TAs or myself deem necessary to contact you about your submitted solutions, you should be available to explain them, otherwise we reserve the right to decrease your score or final grade.

Schedule (subject to change)

Date	Topic	Materials/References
09/25/25	Divide-and-Conquer Karatsuba's integer multiplication Mergesort Master Theorem	
09/30/25	Dynamic Programming	
10/02/25	Dynamic Programming	

10/07/25	Dynamic Programming	
10/09/25	Hashing and its Applications	
10/14/25	Hashing and its Applications	
10/21/25	Linear Programming and Integer Programming	
10/23/25	Linear Programming and Integer Programming	
10/28/25	DFS and the concept of finishing times	
10/30/25	DFS and sorting a DAG using topological sort	
11/04/25	Single source shortest paths and the Bellman Ford algorithm	
11/06/25	Dijkstra's algorithm, All Pairs Shortest Path Problem, the Matrix Multiplication connection	
11/11/25	No lecture (Veterans day)	
11/13/25	Greedy Algorithms: Fractional Knapsack	
11/18/25	Greedy Algorithms: Huffman Coding	
11/20/25	NP-completeness	
11/25/25	Approximation Algorithms	
11/27/25	Online Algorithms	
12/02/25	The Fast Fourier Transform	
12/04/25	All Questions Answered	
12/08/25	Final Exam	

- Other topics to be covered potentially:
 - Hashing (with emphasis on its application and/or universal hashing)
- Assignments and exams will be available on Canvas.

Homework

UC Santa Cruz is committed to creating an academic environment that supports its diverse student body. If you are a student with a disability who requires accommodations to achieve equal access in this course, please affiliate with the DRC. I encourage all students to benefit from learning more about DRC services to contact DRC by phone at 831-459-2089 or by email at drc@ucsc.edu. For students already affiliated, make sure that you have requested Academic Access Letters, where you intend to use accommodations. You can also request to meet privately with me during my office hours or by appointment, as soon as possible. I would like us to discuss how we can implement your accommodations in this course to ensure your access and full engagement in this course

How Homework Works: (Or is it "How Homeworks Work?") There are two kinds of assignments: algorithmic/mathematical problems and coding problems.

First, about algorithmic problems. I will put out a list of questions once every 3-4 lectures. Each homework will have some sections with compulsory questions, and some sections where you can choose a problem. The total will be 15 points, for each homework.

- All solutions must be written in Latex, compiled to a pdf, and uploaded on Canvas.
- Make sure you title your document clearly, specifying the Question document for that solution. Clearly, number and state the problem(s) you have solved.
- Grading takes time! Please be patient.

Now, about coding problems. These involve coding up an algorithm and maybe some wall-clock timing analysis.

- All solutions should be done in a Jupyter Notebook, which should have the code as well as the analysis.
- For grading, we will use the Crowdgrader tool. Collectively, the students will grade each other's solutions. Each student will get five assignments to grade, and less than a week to grade them. (Deadlines will be given on Crowdgrader.) It will help you to see other solutions (both good and bad).
- The TA and myself will supervise the grading. If you feel that your scores are (grossly) unfair, we will regrade your homework.

Deadlines are at 11:59 p.m. on that day. No late submissions will be accepted. You should consider 11:30 p.m. a hard deadline and the 11:30-11:59 p.m. window a grace period. If you cannot submit your work by 11:30 p.m. due to a system crash, you must email your solution to the TAs immediately. It is your responsibility to ensure your submission is successful.

Rules on using outside text: Plagiarism and cheating often happen because students do not understand how to attribute their sources properly. There's a lot of material available, either online or from previous courses. Yes, you are allowed to read the results of Generative AI tools like Github Copilot and ChatGPT. But you are not allowed to copy text/code directly from there. You must write your own solutions. Please follow these rules, so that I know you have not cheated. These rules protect you from allegations of cheating.

- If you use text from anywhere, you must cite that in your submission.
- If you get text from a website, you must cite that in your submission.
- If you used Generative AI tools, state that clearly and explain how you used it.

- You can discuss the assignment with other students, but write your own answers. In general, write your solutions 30 minutes after a discussion, to ensure that you do not write the same text as someone else.
- You must be able to explain **every** sentence that you've written.

FAQ

- To be added